

Notes on Lagrange-Multiplier Prediction for OmniRetargeting

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1 From DDE-Based Retargeting to Multiplier Prediction

These notes reorganize the online OmniRetargeting formulation around a simple computational observation. The robot configuration can be advanced by one forward step if the appropriate Lagrange multipliers are available. The main prediction problem is therefore not necessarily to re-solve the entire primal optimization problem at every frame, but to predict the multiplier pair that enforces the constraints during that step. This section summarizes the viewpoint; the two resulting algorithms are developed in the following sections of the full notes.

1.1 Online constrained retargeting

Let

$$\mathcal{M} := \mathbb{S}^3 \times \mathbb{R}^d, \quad \mathbf{q}_k = (\mathbf{r}_k, \mathbf{x}_k) \in \mathcal{M}, \quad \mathbf{r}_k \in \mathbb{S}^3, \quad \mathbf{x}_k \in \mathbb{R}^d, \quad (1)$$

denote the retargeted configuration at frame k . The unit quaternion $\mathbf{r}_k$ represents the floating-base orientation, while $\mathbf{x}_k$ contains the remaining Euclidean variables, such as the base translation and joint angles. Configuration restrictions such as joint bounds are included in the constraint maps below.

We use the product-manifold derivative notation of the main manuscript. For a scalar function $\psi : \mathcal{M} \rightarrow \mathbb{R}$, define

$$\mathcal{P}_{\mathbf{r}} := I_4 - \mathbf{r}\mathbf{r}^\top, \quad \text{grad}_{\mathbf{r}} \psi(\mathbf{r}, \mathbf{x}) := \mathcal{P}_{\mathbf{r}} \nabla_{\mathbf{r}} \psi(\mathbf{r}, \mathbf{x}), \quad (2)$$

and

$$\mathfrak{D}_{\mathbf{q}} \psi(\mathbf{r}, \mathbf{x}) := \begin{bmatrix} \text{grad}_{\mathbf{r}} \psi(\mathbf{r}, \mathbf{x}) \\ \nabla_{\mathbf{x}} \psi(\mathbf{r}, \mathbf{x}) \end{bmatrix}. \quad (3)$$

Thus $\nabla_{\mathbf{r}}$ is the ambient quaternion derivative, $\text{grad}_{\mathbf{r}}$ is its projection onto $T_{\mathbf{r}}\mathbb{S}^3$, and $\nabla_{\mathbf{x}}$ is the ordinary Euclidean gradient. For a vector-valued map $c : \mathcal{M} \rightarrow \mathbb{R}^s$, the convention is understood row-wise:

$$\text{grad}_{\mathbf{r}} c(\mathbf{q}) := \nabla_{\mathbf{r}} c(\mathbf{q}) \mathcal{P}_{\mathbf{r}}, \quad \mathfrak{D}_{\mathbf{q}} c(\mathbf{q})[d] := \text{grad}_{\mathbf{r}} c(\mathbf{q}) d_r + \nabla_{\mathbf{x}} c(\mathbf{q}) d_x, \quad (4)$$

where $d = (d_r, d_x) \in T_{\mathbf{r}}\mathbb{S}^3 \times \mathbb{R}^d$. Its adjoint is

$$\mathfrak{D}_{\mathbf{q}} c(\mathbf{q})^\top \eta = \begin{bmatrix} \text{grad}_{\mathbf{r}} c(\mathbf{q})^\top \eta \\ \nabla_{\mathbf{x}} c(\mathbf{q})^\top \eta \end{bmatrix}. \quad (5)$$

At frame k , the human/object observation and its relational graph are represented by data P_k and L_k . A typical graph-based retargeting loss is

$$f_k(\mathbf{r}, \mathbf{x}) := \|L_k \phi(\mathbf{r}, \mathbf{x}) - L_k P_k\|_F^2, \quad (6)$$

where ϕ is the robot forward-kinematics map. The exact details of the graph are not needed for the present discussion; what matters is that f_k changes as the input pose changes.

The online update is modeled as a sequence of proximal constrained problems. Given the previous retargeted state $\mathbf{q}_k^*$, the next state is

$$\begin{aligned} \mathbf{q}_{k+1}^* &\in \arg \min_{\mathbf{r}, \mathbf{x}} f_{k+1}(\mathbf{r}, \mathbf{x}) + \frac{1}{2\delta} \left(\|\mathbf{r} - \mathbf{r}_k^*\|^2 + \|\mathbf{x} - \mathbf{x}_k^*\|^2 \right) \\ \text{subject to } &g_{k+1}(\mathbf{r}, \mathbf{x}) \leq 0, \quad h_{k+1}(\mathbf{r}, \mathbf{x}) = 0, \quad \mathbf{r} \in \mathbb{S}^3. \end{aligned} \quad (7)$$

Here g_{k+1} collects inequality constraints (for example, collision, contact, displacement, velocity, and joint-limit constraints), and h_{k+1} collects equality constraints. Both the objective and the constraint maps may change with the frame data. The quadratic term is a short-term tracking or proximal term, and $\delta > 0$ is the frame step size. On the quaternion factor, the displayed Euclidean norm is the chordal local model; the corresponding stationarity equation is understood on the tangent space of $\mathbb{S}^3$.

For the inequality convention $g_k \leq 0$, define

$$\mathcal{L}_k(\mathbf{r}, \mathbf{x}; \lambda, \mu) = f_k(\mathbf{r}, \mathbf{x}) + \lambda^\top g_k(\mathbf{r}, \mathbf{x}) + \mu^\top h_k(\mathbf{r}, \mathbf{x}), \quad \lambda \geq 0. \quad (8)$$

The multiplier λ is associated with inequalities and μ with equalities. For an inequality component, the KKT conditions are

$$\lambda_i \geq 0, \quad g_{k,i}(\mathbf{r}, \mathbf{x}) \leq 0, \quad \lambda_i g_{k,i}(\mathbf{r}, \mathbf{x}) = 0. \quad (9)$$

Thus a positive multiplier is a constraint reaction, whereas a strictly feasible constraint has zero multiplier.

1.2 The implicit KKT update

Let $(\mathbf{q}_{k+1}^*, \lambda_{k+1}^*, \mu_{k+1}^*)$ be a regular KKT solution of (7). The Euclidean stationarity equation is

$$\frac{1}{\delta}(\mathbf{x}_{k+1}^* - \mathbf{x}_k^*) + \nabla_{\mathbf{x}} \mathcal{L}_{k+1}(\mathbf{r}_{k+1}^*, \mathbf{x}_{k+1}^*; \lambda_{k+1}^*, \mu_{k+1}^*) = 0. \quad (10)$$

Consequently,

$$\mathbf{x}_{k+1}^* = \mathbf{x}_k^* - \delta \nabla_{\mathbf{x}} \mathcal{L}_{k+1}(\mathbf{r}_{k+1}^*, \mathbf{x}_{k+1}^*; \lambda_{k+1}^*, \mu_{k+1}^*). \quad (11)$$

The quaternion stationarity equation has the same structure after projection onto the tangent space:

$$\frac{1}{\delta} \mathcal{P}_{\mathbf{r}_{k+1}^*}(\mathbf{r}_{k+1}^* - \mathbf{r}_k^*) + \text{grad}_{\mathbf{r}} \mathcal{L}_{k+1}(\mathbf{r}_{k+1}^*, \mathbf{x}_{k+1}^*; \lambda_{k+1}^*, \mu_{k+1}^*) = 0. \quad (12)$$

If $v_{r,k+1}^* = \text{grad}_{\mathbf{r}} \mathcal{L}_{k+1}$ denotes the tangent gradient evaluated at the new state, then the stationarity equation is equivalently the forward TaSP relation based at the unknown new quaternion,

$$\mathbf{r}_k^* = \text{TaSP}_{\mathbf{r}_{k+1}^*}(\delta v_{r,k+1}^*) = \sqrt{1 - \delta^2 \|v_{r,k+1}^*\|^2} \mathbf{r}_{k+1}^* + \delta v_{r,k+1}^*. \quad (13)$$

Solving this relation for $\mathbf{r}_{k+1}^*$ gives

$$\mathbf{r}_{k+1}^* = \frac{\mathbf{r}_k^* - \delta v_{r,k+1}^*}{\sqrt{1 - \delta^2 \|v_{r,k+1}^*\|^2}}, \quad (14)$$

whenever the denominator is positive. Thus the fraction is an implicit inversion of the forward TaSP relation, not the forward TaSP definition itself. Equations (11)–(14) are *implicit*: both the new configuration and its multipliers appear inside the Lagrangian field on the right-hand side. Therefore, knowing the old configuration alone does not produce the new state by one gradient evaluation. One must solve the new-frame KKT equations, or an SQP approximation of them, to obtain the quantities on the right.

1.3 The previous DDE-based viewpoint

The earlier formulation represented the online KKT evolution by a continuous-time product-manifold flow. Writing τ for the algorithmic time of this solver and $\mathfrak{D}_{\mathbf{q}}$ for the product gradient, the formal flow is

$$\dot{\mathbf{q}}(\tau) = -\mathfrak{D}_{\mathbf{q}} \mathcal{L}(\mathbf{q}(\tau), \lambda^*(\tau), \mu^*(\tau)). \quad (15)$$

The multipliers in this equation are not freely chosen controls. They are determined by the KKT conditions, which involve both the current configuration and the previous configuration through the proximal term. Locally, this dependence can be denoted by

$$(\lambda^*(\tau), \mu^*(\tau)) = \mathcal{M}(\mathbf{q}(\tau), \mathbf{q}(\tau - \delta); \text{frame data}). \quad (16)$$

Substitution gives the delay differential equation (DDE)

$$\dot{\mathbf{q}}(\tau) = -\mathfrak{D}_{\mathbf{q}} \mathcal{L}(\mathbf{q}(\tau), \mathcal{M}(\mathbf{q}(\tau), \mathbf{q}(\tau - \delta); \text{frame data})). \quad (17)$$

This DDE is a useful continuous-time description of the constrained retargeting dynamics and of the delay introduced by frame-to-frame proximal tracking. In computation, however, it must be numerically advanced or relaxed. Reaching an accurate frame solution generally requires multiple integration steps, residual-reduction steps, or SQP iterations. The DDE therefore supplies a principled solver and an error-analysis framework, but it does not by itself remove the inner iterations needed at each incoming frame.

1.4 Reframing the online task as multiplier prediction

The implicit equations reveal where the difficulty is concentrated. If the correct multipliers were known, the primal variables could be advanced directly. For a forward method, all quantities are evaluated at the currently available frame- k state and data. Let

$$\hat{\nu}_{k+1|k} := \begin{bmatrix} \hat{\lambda}_{k+1|k} \\ \hat{\mu}_{k+1|k} \end{bmatrix} \quad (18)$$

be a prediction of the next-frame KKT multipliers. Define

$$v_{r,k} := \text{grad}_{\mathbf{r}} \mathcal{L}_k(\mathbf{r}_k, \mathbf{x}_k; \hat{\lambda}_{k+1|k}, \hat{\mu}_{k+1|k}), \quad v_{x,k} := \nabla_{\mathbf{x}} \mathcal{L}_k(\mathbf{r}_k, \mathbf{x}_k; \hat{\lambda}_{k+1|k}, \hat{\mu}_{k+1|k}). \quad (19)$$

The explicit one-step update is then

$$\mathbf{r}_{k+1} = \text{TaSP}_{\mathbf{r}_k}(-\delta v_{r,k}) = \sqrt{1 - \delta^2 \|v_{r,k}\|^2} \mathbf{r}_k - \delta v_{r,k}, \quad \mathbf{x}_{k+1} = \mathbf{x}_k - \delta v_{x,k}. \quad (20)$$

Here $v_{r,k} \in T_{\mathbf{r}_k} \mathbb{S}^3$ and $\delta \|v_{r,k}\| \leq 1$, so the quaternion expression is the forward TaSP map based at the known current quaternion. Most importantly, (20) uses $\mathcal{L}_k$, not the unknown new-frame field $\mathcal{L}_{k+1}$. The only missing information is the constraint reaction encoded by $\hat{\nu}_{k+1|k}$.

This reframing does not claim that the multipliers are arbitrary or independent states. They are constrained by primal feasibility, dual feasibility, and complementarity. In particular, predicting λ also means predicting which inequality constraints are active. A positive predicted multiplier corresponds to a constraint that blocks the forward motion; a zero multiplier corresponds to a constraint that does not need to react at that step. This is why active set switching is central to the multiplier-prediction problem.

Under a regular fixed active set, the local multiplier map is smooth and the multiplier prediction can be analyzed by differentiating the KKT system. At switching points, the map is only piecewise smooth and the complementarity mechanism must select the new active set. The two methods below address this issue in different ways.

1.5 Two routes for predicting the multipliers

Route A: one-step normal-cone projected dynamics. Begin with the unconstrained forward gradient and project it against the negative normal cone of the feasible set. With $a_k := \mathfrak{D}_{\mathbf{q}} f_k(\mathbf{q}_k)$, define

$$p_k = \Pi_{-N_{C_k}(\mathbf{q}_k)}(a_k), \quad d_k = -a_k + p_k.$$

The configuration is then advanced exactly once by forward TaSP in the quaternion component and by forward Euler in the Euclidean component. This route states the constraint correction geometrically; it does not introduce a multiplier ODE or prescribe an algebraic solver for the projection.

Route B: one-step algebraic multiplier method. Compute the unconstrained forward direction, linearize the current constraints over one step, and find the feasible direction closest to the unconstrained one. With d_k denoting this single direction, the local problem has the form

$$\begin{aligned} \min_d \quad & \frac{\delta}{2} \|d - d_k^0\|^2 \\ \text{subject to} \quad & g_k(\mathbf{q}_k) + \delta G_k[d] \leq 0, \quad h_k(\mathbf{q}_k) + \delta H_k[d] = 0, \end{aligned} \quad (21)$$

where d_k^0 is the unconstrained Euler direction. The KKT multiplier of this small quadratic program is the algebraic prediction $\hat{\nu}_{k+1|k}$, and its complementarity conditions choose the active set automatically. The primal variables are then updated once using (20).

The two routes emphasize different parts of the same one-step correction. The normal-cone formulation specifies the projected primal direction geometrically. The algebraic method developed later supplies a concrete local direction problem and exposes the associated constraint multipliers. Under smoothness, LICQ, and a nondegenerate fixed active set, the latter typically gives an $\mathcal{O}(\delta)$ multiplier prediction error. Since a multiplier enters the primal update through the factor δ , this produces an $\mathcal{O}(\delta^2)$ local primal error and, over a fixed time horizon, an $\mathcal{O}(\delta)$ global primal tracking error. Active-set switching requires a separate local argument, but finitely many nondegenerate switches are compatible with the same first-order global target.

2 Normal-Cone Projection

This section constructs the constrained update from the unconstrained one. Without constraints, the proximal problem gives a backward Euler relation, and one forward gradient evaluation gives its explicit approximation. With constraints, the same forward direction is corrected by a projection onto the negative normal cone. The corrected direction is then applied once by TaSP in the quaternion component and by forward Euler in the Euclidean component. No multiplier differential equation and no inner primal–dual iteration are introduced.

2.1 The unconstrained backward and forward steps

Let

$$\mathbf{q}_k = (\mathbf{r}_k, \mathbf{x}_k) \in \mathcal{M}, \quad \mathcal{M} = \mathbb{S}^3 \times \mathbb{R}^d. \quad (22)$$

Ignoring the constraints, consider the proximal update

$$\mathbf{q}_{k+1}^B \in \arg \min_{\mathbf{q} \in \mathcal{M}} \left\{ f_{k+1}(\mathbf{q}) + \frac{1}{2\delta} \|\mathbf{q} - \mathbf{q}_k\|^2 \right\}. \quad (23)$$

For any scalar function $\psi(\mathbf{r}, \mathbf{x})$, use the same product-gradient convention as in the main manuscript:

$$\mathfrak{D}_{\mathbf{q}} \psi(\mathbf{r}, \mathbf{x}) := \begin{bmatrix} \text{grad}_{\mathbf{r}} \psi(\mathbf{r}, \mathbf{x}) \\ \nabla_{\mathbf{x}} \psi(\mathbf{r}, \mathbf{x}) \end{bmatrix}, \quad \text{grad}_{\mathbf{r}} \psi(\mathbf{r}, \mathbf{x}) := \mathcal{P}_{\mathbf{r}} \nabla_{\mathbf{r}} \psi(\mathbf{r}, \mathbf{x}), \quad \mathcal{P}_{\mathbf{r}} := I_4 - \mathbf{r} \mathbf{r}^\top. \quad (24)$$

Thus $\mathfrak{D}_{\mathbf{q}}$ is not an ambient Euclidean gradient: its quaternion block is Riemannian, whereas its $\mathbf{x}$ -block is Euclidean. Writing $\mathbf{q}_{k+1}^B = (\mathbf{r}_{k+1}^B, \mathbf{x}_{k+1}^B)$, the exact chordal stationarity equation is

$$\begin{bmatrix} \frac{1}{\delta} \mathcal{P}_{\mathbf{r}_{k+1}^B} (\mathbf{r}_{k+1}^B - \mathbf{r}_k) + \text{grad}_{\mathbf{r}} f_{k+1}(\mathbf{r}_{k+1}^B, \mathbf{x}_{k+1}^B) \\ \frac{1}{\delta} (\mathbf{x}_{k+1}^B - \mathbf{x}_k) + \nabla_{\mathbf{x}} f_{k+1}(\mathbf{r}_{k+1}^B, \mathbf{x}_{k+1}^B) \end{bmatrix} = \mathbf{0}. \quad (25)$$

The unknown new state appears inside the gradient, so (25) is a backward Euler relation. To recover the quaternion formula used in the main manuscript, define

$$v_{r,k+1}^B := \text{grad}_{\mathbf{r}} f_{k+1}(\mathbf{r}_{k+1}^B, \mathbf{x}_{k+1}^B) \in T_{\mathbf{r}_{k+1}^B} \mathbb{S}^3. \quad (26)$$

For $\mathbf{r}_0 \in \mathbb{S}^3$ and $v \in T_{\mathbf{r}_0} \mathbb{S}^3$ with $\|v\| \leq 1$, the forward TaSP map is

$$\boxed{\text{TaSP}_{\mathbf{r}_0}(v) := \sqrt{1 - \|v\|^2} \mathbf{r}_0 + v.} \quad (27)$$

Because $v_{r,k+1}^B$ is tangent at the unknown new quaternion, the quaternion block of (25) can first be written as

$$\mathbf{r}_k = \text{TaSP}_{\mathbf{r}_{k+1}^B}(\delta v_{r,k+1}^B) = \sqrt{1 - \delta^2 \|v_{r,k+1}^B\|^2} \mathbf{r}_{k+1}^B + \delta v_{r,k+1}^B. \quad (28)$$

Solving this forward TaSP relation for the unknown new quaternion gives the implicit dynamics used in the main manuscript:

$$\boxed{\mathbf{r}_{k+1}^B = \frac{\mathbf{r}_k - \delta v_{r,k+1}^B}{\sqrt{1 - \delta^2 \|v_{r,k+1}^B\|^2}}}. \quad (29)$$

Thus the fraction in (29) is not the definition of forward TaSP. It is the result of inverting a TaSP relation whose base point is the unknown $\mathbf{r}_{k+1}^B$.

The corresponding forward approximation evaluates the available field at the current state. Define

$$a_k := \mathfrak{D}_{\mathbf{q}} f_k(\mathbf{q}_k) = \begin{bmatrix} a_{r,k} \\ a_{x,k} \end{bmatrix}, \quad a_{r,k} = \mathcal{P}_{\mathbf{r}_k} \nabla_{\mathbf{r}} f_k(\mathbf{r}_k, \mathbf{x}_k), \quad a_{x,k} = \nabla_{\mathbf{x}} f_k(\mathbf{r}_k, \mathbf{x}_k), \quad (30)$$

where, for the OmniRetargeting loss used in the main manuscript,

$$f_k(\mathbf{r}, \mathbf{x}) = \|E_k(\mathbf{r}, \mathbf{x})\|_F^2, \quad E_k(\mathbf{r}, \mathbf{x}) := L_k \phi(\mathbf{r}, \mathbf{x}) - L_k P_k. \quad (31)$$

Now $a_{r,k} \in T_{\mathbf{r}_k} \mathbb{S}^3$, so TaSP can be applied directly at the known current quaternion. Provided $\delta \|a_{r,k}\| \leq 1$, the unconstrained one-step forward approximation is

$$\mathbf{r}_{k+1}^F = \text{TaSP}_{\mathbf{r}_k}(-\delta a_{r,k}) = \sqrt{1 - \delta^2 \|a_{r,k}\|^2} \mathbf{r}_k - \delta a_{r,k}, \quad \mathbf{x}_{k+1}^F = \mathbf{x}_k - \delta a_{x,k}. \quad (32)$$

This formula is genuinely forward: both its vector field and tangent space are evaluated at $\mathbf{r}_k$. The square-root coefficient and the orthogonal tangent component have squared norms summing to one. Under the usual local smoothness assumptions, this forward step differs from the backward step by $\mathcal{O}(\delta^2)$ over one frame.

2.2 Adding the constraint set

Let the frame- k feasible set be

$$\mathcal{C}_k := \{(\mathbf{r}, \mathbf{x}) \in \mathcal{M} : g_k(\mathbf{r}, \mathbf{x}) \leq 0, \quad h_k(\mathbf{r}, \mathbf{x}) = 0\}. \quad (33)$$

The constrained proximal update is described by the backward inclusion

$$0 \in \begin{bmatrix} \frac{1}{\delta} \mathcal{P}_{\mathbf{r}_{k+1}^B}(\mathbf{r}_{k+1}^B - \mathbf{r}_k) \\ \frac{1}{\delta}(\mathbf{x}_{k+1}^B - \mathbf{x}_k) \end{bmatrix} + \mathfrak{D}_{\mathbf{q}} f_{k+1}(\mathbf{q}_{k+1}^B) + N_{\mathcal{C}_{k+1}}^{\mathcal{M}}(\mathbf{q}_{k+1}^B), \quad (34)$$

where $N_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q})$ is the normal cone of the feasible set relative to the product manifold.

To obtain a one-step explicit approximation, evaluate both the gradient and the constraint geometry at the current point. Project the current gradient onto the negative normal cone:

$$p_k := \Pi_{-N_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q}_k)}(a_k). \quad (35)$$

The actual update direction is the remaining negative-gradient component,

$$\boxed{d_k = -a_k + p_k}. \quad (36)$$

This formula agrees with the Lagrangian-gradient convention in the main manuscript. There, the product field has the form

$$\mathfrak{D}_{\mathbf{q}} \mathcal{L}_k = a_k + n_k, \quad n_k \in N_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q}_k). \quad (37)$$

The normal-cone projection selects $n_k = -p_k$. Hence the corrected Lagrangian field is $a_k - p_k$, and its negative is exactly $d_k = -a_k + p_k$. The projection therefore replaces the unknown constraint-reaction part of the Lagrangian gradient; it does not project the full Lagrangian gradient a second time. Equivalently,

$$d_k = \Pi_{T_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q}_k)}(-a_k), \quad (38)$$

so d_k is the part of the negative gradient that belongs to the feasible tangent cone. At a constrained stationary point, $a_k \in -N_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q}_k)$; hence $p_k = a_k$ and $d_k = 0$. Notice that p_k alone is not the update direction.

2.3 Active geometry and one TaSP–Euler step

At a feasible point, the active inequality set is

$$\mathcal{A}_k := \{i : g_{k,i}(\mathbf{q}_k) = 0\}. \quad (39)$$

Together with the equality constraints, these active constraints determine the local normal cone:

$$N_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q}_k) = \text{cone} \{ \mathfrak{D}_{\mathbf{q}} g_{k,i}(\mathbf{q}_k) : i \in \mathcal{A}_k \} + \text{span} \{ \mathfrak{D}_{\mathbf{q}} h_{k,\ell}(\mathbf{q}_k) : \ell = 1, \dots, p \}. \quad (40)$$

No procedure for computing the projection is specified in this section; only its geometric role is used.

Split the projected direction in (36) as

$$d_k = \begin{bmatrix} d_{r,k} \\ d_{x,k} \end{bmatrix}, \quad d_{r,k} \in T_{\mathbf{r}_k} \mathbb{S}^3. \quad (41)$$

Provided $\delta \|d_{r,k}\| \leq 1$, exactly one primal update is then made:

$$\boxed{\begin{aligned} \mathbf{r}_{k+1}^{\text{F}} &= \text{TaSP}_{\mathbf{r}_k}(\delta d_{r,k}) = \sqrt{1 - \delta^2 \|d_{r,k}\|^2} \mathbf{r}_k + \delta d_{r,k}, \\ \mathbf{x}_{k+1}^{\text{F}} &= \mathbf{x}_k + \delta d_{x,k}. \end{aligned}} \quad (42)$$

The complete construction is therefore

$$\boxed{\begin{aligned} \text{check the active set} &\longrightarrow p_k = \Pi_{-N_{\mathcal{C}_k}^{\mathcal{M}}(\mathbf{q}_k)}(a_k) \longrightarrow d_k = -a_k + p_k \\ &\longrightarrow \text{one forward TaSP/Euler step.} \end{aligned}} \quad (43)$$

Thus the constrained construction is the direct extension of the unconstrained forward step: the negative-gradient motion is kept, while its outward normal component is removed before the configuration is advanced once.

2.4 Assumptions and one-step consistency

The local error statement concerns one smooth branch of the constrained problem. We use the following assumptions.

- (A1) The objective and constraint functions are C^2 in $\mathbf{q}$ on a neighborhood of the step, with uniformly bounded first and second derivatives. The frame-to-frame change of the objective gradient is $\mathcal{O}(\delta)$.
- (A2) The current state is feasible. In the neighborhood used for the comparison, the active inequalities and equalities define the same smooth local constraint geometry. The inactive inequalities have a uniform slack, the active multipliers satisfy strict complementarity, and the active inequality and equality gradients satisfy LICQ.

- (A3) The constrained proximal problem has a locally unique regular KKT branch $\mathbf{q}_{k+1}^B$ satisfying $\|\mathbf{q}_{k+1}^B - \mathbf{q}_k\| = \mathcal{O}(\delta)$. The constraint set is locally fixed during the step. Time-dependent constraint drift is not included in the statement below.
- (A4) The quaternion forward map is TaSP as defined in (27), and the step is small enough that $\delta\|v\| \leq 1$. In one quaternion hemisphere it satisfies

$$\text{TaSP}_{\mathbf{r}}(\delta v) = \mathbf{r} + \delta v + \mathcal{O}(\delta^2) \quad \text{for } v \in T_{\mathbf{r}}\mathbb{S}^3.$$

Assumption (A2) is needed only for the smooth local comparison below. It is not required merely to define the normal-cone projection. At a degenerate switch the projection may remain well defined, but the smooth argument used in the theorem no longer applies directly.

Theorem 2.1 (One-step forward-backward discrepancy). *Suppose (A1)–(A4) hold. Let $\mathbf{q}_{k+1}^B = (\mathbf{r}_{k+1}^B, \mathbf{x}_{k+1}^B)$ be the regular KKT solution of the constrained proximal problem (34). Let $\mathbf{q}_{k+1}^F = (\mathbf{r}_{k+1}^F, \mathbf{x}_{k+1}^F)$ be the one-step TaSP–Euler update obtained from (35)–(42). Then there exist constants $\bar{\delta} > 0$ and $C > 0$, independent of δ , such that*

$$\|\mathbf{r}_{k+1}^B - \mathbf{r}_{k+1}^F\| + \|\mathbf{x}_{k+1}^B - \mathbf{x}_{k+1}^F\| \leq C\delta^2, \quad 0 < \delta \leq \bar{\delta}, \quad (44)$$

where the two quaternions are represented in the same hemisphere. Moreover, the tangent-cone projected step has a nonlinear feasibility residual of the same order:

$$\|h_k(\mathbf{q}_{k+1}^F)\| + \|[g_k(\mathbf{q}_{k+1}^F)]_+\| \leq C\delta^2. \quad (45)$$

Proof. Work in a smooth local product chart centered at $\mathbf{q}_k$, and denote the projected vector field by

$$V_k(\mathbf{q}) := \Pi_{T_{C_k}^{\mathcal{M}}(\mathbf{q})}(-\mathfrak{D}_{\mathbf{q}}f_k(\mathbf{q})).$$

Under (A1)–(A3), the active geometry is fixed and regular, so V_k is locally Lipschitz. In the local chart, the constrained backward stationarity relation and the one-step forward update have the forms

$$\xi_{k+1}^B = \delta V_{k+1}(\xi_{k+1}^B) + \mathcal{O}(\delta^2), \quad \xi_{k+1}^F = \delta V_k(0) + \mathcal{O}(\delta^2).$$

The first relation and (A3) give $\|\xi_{k+1}^B\| = \mathcal{O}(\delta)$. Local Lipschitz continuity and the $\mathcal{O}(\delta)$ frame variation then imply

$$\delta\|V_{k+1}(\xi_{k+1}^B) - V_k(0)\| = \mathcal{O}(\delta^2).$$

Subtracting the two relations proves (44); (A4) transfers the chart estimate to the TaSP and Euclidean components.

Finally, d_k is tangent to the active inequalities and equalities. Since the one-step displacement is $\mathcal{O}(\delta)$, Taylor expansion of g_k and h_k along this direction leaves only $\mathcal{O}(\delta^2)$ remainders, which gives (45). $\square$

Theorem 2.1 is a one-step consistency result. It does not require global strong convexity or a PL condition. Such conditions become relevant only when one asks whether repeated forward steps converge or track a moving solution over many frames.

3 One-Step Algebraic Multiplier Method

Section 2 introduced the constraint correction geometrically as a projection onto a negative normal cone. This section develops an algebraic construction of that correction and studies its relation to the next-frame KKT multipliers. The local problem computes the constraint reaction for the current frame transition, while the configuration itself is advanced only once.

3.1 A scalar wall example

It is useful to begin with one scalar variable and one inequality. Let the feasible set be the half-line

$$x \leq 0, \quad (46)$$

and let f_k be the loss at the current frame. Without the constraint, one forward Euler step would use the direction

$$d_k^0 = -f'_k(x_k), \quad \tilde{x}_{k+1} = x_k + \delta d_k^0. \quad (47)$$

If $\tilde{x}_{k+1} \leq 0$, this step is already feasible and no constraint reaction is required. If $\tilde{x}_{k+1} > 0$, the step would cross the wall. We then choose the closest feasible direction:

$$\begin{aligned} d_k \in \arg \min_d \quad & \frac{\delta}{2} |d - d_k^0|^2 \\ \text{subject to} \quad & x_k + \delta d \leq 0. \end{aligned} \quad (48)$$

The solution is simply

$$d_k = \min \left\{ d_k^0, -\frac{x_k}{\delta} \right\}, \quad x_{k+1} = x_k + \delta d_k. \quad (49)$$

Let $\hat{\lambda}_{k+1|k}$ denote the multiplier of (48). Its stationarity condition is

$$\delta(d_k - d_k^0) + \delta \hat{\lambda}_{k+1|k} = 0, \quad (50)$$

and hence

$$d_k = -(f'_k(x_k) + \hat{\lambda}_{k+1|k}). \quad (51)$$

The multiplier has the explicit form

$$\boxed{\hat{\lambda}_{k+1|k} = \max \left\{ 0, \frac{x_k}{\delta} - f'_k(x_k) \right\}.} \quad (52)$$

Thus $\hat{\lambda}_{k+1|k}$ is the smallest nonnegative reaction that prevents the current forward step from crossing the wall. Notice the switching behavior. If the free step points into the feasible half-line, then $\hat{\lambda}_{k+1|k} = 0$. If it points through the wall, then $\hat{\lambda}_{k+1|k} > 0$ and the corrected step ends at $x_{k+1} = 0$. At a later frame, if the free direction points back into the interior, the multiplier returns to zero. The constraint therefore does not remain active merely because it was active at the preceding frame.

3.2 The proposed one-step direction

Following the scalar wall example, we now propose the corresponding one-step direction for OmniRetargeting. The current gradient is

$$a_k := \mathfrak{D}_{\mathbf{q}} f_k(\mathbf{q}_k) = \begin{bmatrix} \text{grad}_{\mathbf{r}} f_k(\mathbf{q}_k) \\ \nabla_{\mathbf{x}} f_k(\mathbf{q}_k) \end{bmatrix} \in T_{\mathbf{r}_k} \mathbb{S}^3 \times \mathbb{R}^d. \quad (53)$$

Define the linearized constraint operators $G_k : T_{\mathbf{r}_k} \mathbb{S}^3 \times \mathbb{R}^d \rightarrow \mathbb{R}^m$ and $H_k : T_{\mathbf{r}_k} \mathbb{S}^3 \times \mathbb{R}^d \rightarrow \mathbb{R}^p$. For the vector-valued constraint maps, use the row-wise Riemannian Jacobians

$$\text{grad}_{\mathbf{r}} g_k(\mathbf{q}_k) := \nabla_{\mathbf{r}} g_k(\mathbf{q}_k) \mathcal{P}_{\mathbf{r}_k}, \quad \text{grad}_{\mathbf{r}} h_k(\mathbf{q}_k) := \nabla_{\mathbf{r}} h_k(\mathbf{q}_k) \mathcal{P}_{\mathbf{r}_k}. \quad (54)$$

Thus each row is the transpose of the Riemannian gradient of one constraint component. The two linearized operators are

$$\begin{aligned} G_k[d] &:= \mathfrak{D}_{\mathbf{q}} g_k(\mathbf{q}_k)[d] = \text{grad}_{\mathbf{r}} g_k(\mathbf{q}_k) d_r + \nabla_{\mathbf{x}} g_k(\mathbf{q}_k) d_x, \\ H_k[d] &:= \mathfrak{D}_{\mathbf{q}} h_k(\mathbf{q}_k)[d] = \text{grad}_{\mathbf{r}} h_k(\mathbf{q}_k) d_r + \nabla_{\mathbf{x}} h_k(\mathbf{q}_k) d_x, \\ d &:= \begin{bmatrix} d_r \\ d_x \end{bmatrix} \in T_{\mathbf{r}_k} \mathbb{S}^3 \times \mathbb{R}^d. \end{aligned} \quad (55)$$

The unconstrained forward Euler direction is

$$d_k^0 = -a_k. \quad (56)$$

Following the same idea as in (48), we propose d_k to be the feasible local direction closest to the unconstrained Euler direction d_k^0 . Linearizing the constraints at the current state gives

$$\boxed{\begin{aligned} d_k &\in \arg \min_{d \in T_{\mathbf{r}_k} \mathbb{S}^3 \times \mathbb{R}^d} \frac{\delta}{2} \|d - d_k^0\|^2 \\ &\text{subject to } g_k(\mathbf{q}_k) + \delta G_k[d] \leq 0, \\ &\quad h_k(\mathbf{q}_k) + \delta H_k[d] = 0. \end{aligned}} \quad (57)$$

The objective in (57) has a simple geometric meaning: it projects the desired, unconstrained Euler direction onto the linearized feasible direction set. If d_k^0 already satisfies the linearized constraints, then the projection leaves it unchanged and $d_k = d_k^0$. Otherwise, the QP returns the smallest correction of d_k^0 that restores linearized feasibility. Once $d_k = (d_{r,k}, d_{x,k})$ has been obtained, the actual displacement is δd_k and is applied exactly once. Provided $\delta \|d_{r,k}\| \leq 1$, the update is

$$\boxed{\mathbf{r}_{k+1} = \text{TaSP}_{\mathbf{r}_k}(\delta d_{r,k}) = \sqrt{1 - \delta^2 \|d_{r,k}\|^2} \mathbf{r}_k + \delta d_{r,k}, \quad \mathbf{x}_{k+1} = \mathbf{x}_k + \delta d_{x,k}.} \quad (58)$$

Thus the QP determines the single forward direction; it does not introduce an inner iteration over the original nonlinear retargeting problem.

3.3 KKT conditions and the forward Lagrangian step

Let

$$\widehat{\nu}_{k+1|k} := \begin{bmatrix} \widehat{\lambda}_{k+1|k} \\ \widehat{\mu}_{k+1|k} \end{bmatrix}, \quad \widehat{\lambda}_{k+1|k} \in \mathbb{R}_+^m, \quad \widehat{\mu}_{k+1|k} \in \mathbb{R}^p, \quad (59)$$

denote the inequality and equality multipliers of (57). Let G_k^* and H_k^* be the adjoints of the two tangent-space constraint operators. Explicitly,

$$\begin{aligned} G_k^* \lambda &:= \mathfrak{D}_{\mathbf{q}} g_k(\mathbf{q}_k)^\top \lambda = \begin{bmatrix} \text{grad}_{\mathbf{r}} g_k(\mathbf{q}_k)^\top \lambda \\ \nabla_{\mathbf{x}} g_k(\mathbf{q}_k)^\top \lambda \end{bmatrix}, \\ H_k^* \mu &:= \mathfrak{D}_{\mathbf{q}} h_k(\mathbf{q}_k)^\top \mu = \begin{bmatrix} \text{grad}_{\mathbf{r}} h_k(\mathbf{q}_k)^\top \mu \\ \nabla_{\mathbf{x}} h_k(\mathbf{q}_k)^\top \mu \end{bmatrix}. \end{aligned} \quad (60)$$

The QP stationarity condition is

$$\delta(d_k - d_k^0) + \delta G_k^* \widehat{\lambda}_{k+1|k} + \delta H_k^* \widehat{\mu}_{k+1|k} = 0. \quad (61)$$

Dividing by δ and substituting $d_k^0 = -a_k$ yields

$$\boxed{d_k = -\left(a_k + G_k^* \widehat{\lambda}_{k+1|k} + H_k^* \widehat{\mu}_{k+1|k}\right)}. \quad (62)$$

Recall the current-frame Lagrangian

$$\mathcal{L}_k(\mathbf{r}, \mathbf{x}; \lambda, \mu) := f_k(\mathbf{r}, \mathbf{x}) + \lambda^\top g_k(\mathbf{r}, \mathbf{x}) + \mu^\top h_k(\mathbf{r}, \mathbf{x}). \quad (63)$$

Therefore (62) is the product-manifold gradient step

$$d_k = -\mathfrak{D}_{\mathbf{q}} \mathcal{L}_k(\mathbf{q}_k; \hat{\lambda}_{k+1|k}, \hat{\mu}_{k+1|k}). \quad (64)$$

Its two blocks are

$$d_{r,k} = -\text{grad}_{\mathbf{r}} \mathcal{L}_k(\mathbf{r}_k, \mathbf{x}_k; \hat{\lambda}_{k+1|k}, \hat{\mu}_{k+1|k}), \quad (65)$$

$$d_{x,k} = -\nabla_{\mathbf{x}} \mathcal{L}_k(\mathbf{r}_k, \mathbf{x}_k; \hat{\lambda}_{k+1|k}, \hat{\mu}_{k+1|k}). \quad (66)$$

The small QP determines the single direction d_k ; it is not an inner optimization loop over the original primal variables.

3.4 Complementarity and automatic active-set switching

For the local QP, define the predicted constraint values

$$\hat{g}_{k+1|k} := g_k(\mathbf{q}_k) + \delta G_k[d_k], \quad \hat{h}_{k+1|k} := h_k(\mathbf{q}_k) + \delta H_k[d_k] = 0. \quad (67)$$

The remaining KKT conditions are

$$\boxed{\hat{\lambda}_{k+1|k} \geq 0, \quad \hat{g}_{k+1|k} \leq 0, \quad \hat{\lambda}_{k+1|k,i} \hat{g}_{k+1|k,i} = 0 \quad \text{for every } i.} \quad (68)$$

Thus the active set is selected by the current local problem:

$$\hat{g}_{k+1|k,i} < 0 \implies \hat{\lambda}_{k+1|k,i} = 0, \quad (69)$$

$$\hat{\lambda}_{k+1|k,i} > 0 \implies \hat{g}_{k+1|k,i} = 0. \quad (70)$$

An inequality enters when the unconstrained direction would violate its linearized bound. It leaves when the corrected direction no longer needs a positive reaction. This is why the method can switch active constraints from one frame to the next without integrating a dual ODE or freezing an active set forever.

The exact mathematical mechanism is a quadratic program, or equivalently an LCP after eliminating the equality-constrained directions. In a numerical active-set implementation, tolerances are needed at the degenerate point $\hat{\lambda}_{k+1|k,i} = 0$ and $\hat{g}_{k+1|k,i} = 0$. A typical hysteresis rule is to declare an inequality entering only when the free-step predictor violates its bound by more than an entry threshold or the computed multiplier exceeds its entry threshold, and to release it only when $\hat{\lambda}_{k+1|k,i} < \varepsilon_\lambda$ and $\hat{g}_{k+1|k,i} < -\varepsilon_g$ for prescribed exit tolerances. Hysteresis is a numerical device for suppressing chatter; the underlying complementarity conditions remain the source of the active-set decision.

3.5 Relation to the true nonlinear KKT multipliers

Recall that $\hat{\nu}_{k+1|k}$ in (59) is the multiplier of the frame- k linearized direction QP. For the original nonlinear proximal problem at frame $k+1$, define

$$\nu_{k+1}^* := \begin{bmatrix} \lambda_{k+1}^* \\ \mu_{k+1}^* \end{bmatrix}. \quad (71)$$

In general,

$$\hat{\nu}_{k+1|k} \neq \nu_{k+1}^*; \quad (72)$$

the first vector is a one-step predictor for the second.

To derive the local prediction order, first freeze the frame- k data and compare the QP with the corresponding nonlinear proximal step. On a fixed active-set interval, let $\mathcal{A}_k$ denote the common active inequality set and define the restricted operator

$$G_{k,\mathcal{A}_k}[d] := (G_k[d])_{\mathcal{A}_k}. \quad (73)$$

Its adjoint is denoted by $G_{k,\mathcal{A}_k}^*$. The inactive multiplier components are zero throughout this fixed-active-set calculation. Choose a smooth product chart $y = \chi_k(\mathbf{q})$ centered at $\mathbf{q}_k$. Here ∇_y and D_y are ordinary derivatives in that chart; they are the coordinate representations of the product derivative $\mathfrak{D}_{\mathbf{q}}$ defined in (3). Let $d_{k+1|k}^*$ be the exact nonlinear direction for the frozen frame- k problem, so that

$$y_{k+1|k}^* = y_k + \delta d_{k+1|k}^*, \quad d_{k+1|k}^* = \mathcal{O}(1). \quad (74)$$

Denote its multipliers by $(\lambda_{k+1|k}^*, \mu_{k+1|k}^*)$, and set

$$\nu_{k+1|k}^* := \begin{bmatrix} \lambda_{k+1|k}^* \\ \mu_{k+1|k}^* \end{bmatrix}. \quad (75)$$

On the active branch, its KKT equations are

$$\begin{aligned} \nabla_y f_k(y_{k+1|k}^*) + d_{k+1|k}^* + D_y g_{k,\mathcal{A}_k}(y_{k+1|k}^*)^\top \lambda_{k+1|k,\mathcal{A}_k}^* \\ + D_y h_k(y_{k+1|k}^*)^\top \mu_{k+1|k}^* = 0, \end{aligned} \quad (76)$$

$$g_{k,\mathcal{A}_k}(y_{k+1|k}^*) = 0, \quad h_k(y_{k+1|k}^*) = 0. \quad (77)$$

Evaluating the current linearized equations at this exact direction gives the residuals

$$\rho_{s,k} := a_k + d_{k+1|k}^* + G_{k,\mathcal{A}_k}^* \lambda_{k+1|k,\mathcal{A}_k}^* + H_k^* \mu_{k+1|k}^* = \mathcal{O}(\delta), \quad (78)$$

$$\rho_{g,k} := g_{k,\mathcal{A}_k}(\mathbf{q}_k) + \delta G_{k,\mathcal{A}_k}[d_{k+1|k}^*] = \mathcal{O}(\delta^2), \quad (79)$$

$$\rho_{h,k} := h_k(\mathbf{q}_k) + \delta H_k[d_{k+1|k}^*] = \mathcal{O}(\delta^2). \quad (80)$$

The first order follows from the $\mathcal{O}(\delta)$ change of the objective and constraint derivatives along the step. The last two are the second-order Taylor remainders of the active constraints.

Use the same indexed notation for the three errors:

$$\begin{aligned} e_{d,k} &:= d_k - d_{k+1|k}^*, \\ e_{\lambda,k} &:= \hat{\lambda}_{k+1|k,\mathcal{A}_k} - \lambda_{k+1|k,\mathcal{A}_k}^*, \\ e_{\mu,k} &:= \hat{\mu}_{k+1|k} - \mu_{k+1|k}^*. \end{aligned} \quad (81)$$

Subtracting the residual equations from the QP KKT equations gives the single active-set system

$$\begin{bmatrix} I & G_{k,\mathcal{A}_k}^* & H_k^* \\ G_{k,\mathcal{A}_k} & 0 & 0 \\ H_k & 0 & 0 \end{bmatrix} \begin{bmatrix} e_{d,k} \\ e_{\lambda,k} \\ e_{\mu,k} \end{bmatrix} = - \begin{bmatrix} \rho_{s,k} \\ \rho_{g,k}/\delta \\ \rho_{h,k}/\delta \end{bmatrix}. \quad (82)$$

Under LICQ, the combined operator $[G_{k,\mathcal{A}_k}; H_k]$ has full row rank, so the KKT matrix in (82) is locally nonsingular. Therefore

$$\boxed{e_{\lambda,k} = \mathcal{O}(\delta), \quad e_{\mu,k} = \mathcal{O}(\delta), \quad e_{d,k} = \mathcal{O}(\delta), \quad \delta e_{d,k} = \mathcal{O}(\delta^2).} \quad (83)$$

Thus the multiplier prediction error is first order, while the error in the actual TaSP–Euler displacement is second order. Since the TaSP map itself differs from its tangent increment by $\mathcal{O}(\delta^2)$, it does not change this order.

Finally, suppose the regular nonlinear KKT branch is Lipschitz in the frame data and that the change from frame k to frame $k + 1$ is $\mathcal{O}(\delta)$. Then

$$\nu_{k+1|k}^* - \nu_{k+1}^* = \mathcal{O}(\delta), \quad \hat{\nu}_{k+1|k} - \nu_{k+1}^* = \mathcal{O}(\delta). \quad (84)$$

Whenever the complete one-step consistency error, including the frame-dependent constraint data, is $\mathcal{O}(\delta^2)$, the state error obeys

$$E_{k+1} \leq (1 + L\delta)E_k + C\delta^2, \quad (85)$$

and hence, over a fixed time horizon,

$$\max_{0 \leq k \leq T/\delta} E_k = \mathcal{O}(\delta). \quad (86)$$

No global strong convexity is required for this local argument; local KKT regularity, LICQ, boundedness, and a stable solution path are the relevant assumptions.

3.6 Switching points and limitations

The preceding error calculation assumes that the active set is fixed in the neighborhood of the step. At a switching point the multiplier map is generally only piecewise smooth, so a fixed-active-set derivative or multiplier ODE cannot be continued blindly through the event. The one-step method handles the event by resolving the complementarity conditions in the current QP:

1. warm-start the QP solver with the preceding frame's multipliers and active set;
2. solve the current linearized direction problem;
3. accept the entering and leaving constraints selected by complementarity, setting released multipliers to zero;
4. apply exactly one TaSP–Euler update with displacement δd_k .

If there are finitely many nondegenerate transversal switches, an $\mathcal{O}(\delta)$ event-location error is compatible with the global $\mathcal{O}(\delta)$ state accuracy. Chattering, nearly dependent active constraints, simultaneous degenerate contacts, or Zeno-type infinitely many switches require additional regularization, a carefully chosen hysteresis rule, or a higher-order event treatment.

Finally, the QP enforces the linearization of the frame- k constraints, not the full nonlinear constraints. The residual with respect to g_k, h_k after one step is generally $\mathcal{O}(\delta^2)$ under smoothness. If the constraint maps change between frames, the drift from g_k, h_k to g_{k+1}, h_{k+1} must be included separately in the predictor or in the consistency estimate. Collision constraints that must never be violated may therefore require a safety margin, a line search, or a nonlinear feasibility correction. Also, in the general coupled case, identifying the active set is itself a complementarity problem; there is no universally valid explicit formula for all multipliers that avoids this local algebraic solve. The main advantage is that this solve produces the multiplier prediction and is followed by only one forward update of $(\mathbf{r}, \mathbf{x})$.